Candidate's Name/ Index Number ……

THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE TERMINAL EXAMINATION
033 BIOLOGY

Time: 2½ Hours **Tuesday, 27th May 2025 p.m.**

INSTRUCTIONS

1. This paper consists of THREE (3) sections A, B, and C with a total of ten (10) questions.
2. Attempt **ALL** questions.
3. All answers must be written in the spaces provided.
4. Write your examination number on every page in this question paper.
5. All writing must be in blue/black ink. Diagrams must be drawn in pencil and labeled in ink.
6. Cell phones and calculators are not allowed in the examination room

QUESTION NUMBER	MARKS	INITIALS
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

Candidate's Name/ Index Number …….

SECTION A (15 MARKS)

1. Select the most correct answer in each item and write its letter beside its corresponding item number.

i. The two main branches of biology are:
   A. Botany and Zoology
   B. Zoology and Bacteriology
   C. Anthropology and Sociology
   D. Anatomy and Cytology

ii. Which cell organelle is responsible for energy production?
   A. Ribosome
   B. Chloroplast
   C. Mitochondrion
   D. Lysosome

iii. Characteristic of living organism where the orgnisms break down food substance to relase eneergy is called,
   A. Reproduction
   B. Nutrition
   C. Irritability
   D. Respiration

iv. The following are not branches of biology except
   A. Sociology
   B. Science
   C. Anthropology
   D. Anatomy

v. Among the following branches of biology, which one is the study of classification of organisms.
   A. Taxonomy
   B. Ecology
   C. Classication
   D. Nomenclature

vi. Organelles responsible for protein synthesis in a cell are called
   A. Centrioles
   B. Chromosomes
   C. Lysosomes
   D. Ribosomes

vii. Which of the followinng structures of organisms are homologous?
   A. Wings of birds and wings of grasshoppers
   B. Wings of bats and arms of baboons
   C. Arms of man and wings of butterfly
   D. Wings of housefly and wings of butterfly

viii. Science subject that studies the structure of plant and animal organs is known as
A. Histology
B. Anatomy
C. Cytology
D. Virology

ix. The word Biology comes from two words: Bios and Logos. The word logos means,
A. Life B. Study of C. Logic D. Animals

x. Charateristic of living organism where the orgnisms responds to a stimulus
A. Reproduction
B. Nutrition
C. Irritability
D. Respiration

Answers

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the item in List A with the correct item in List B by writing its letter below the number of the corresponding item in List A in the table provided.

LIST A	**LIST B**
(i) A thin semi permeable layer, which encloses the animal cell and control the movement of material in and out of the cell. (ii) A semi-fluid material made up of water and dissolved chemical substances, which supports cell organelles. (iii) An oval or spherical organelle suspended in the cytoplasm, which controls the functions of all other parts of the cell. (iv) Oval-shaped organelles with double membranes, which produce energy. (v) Fluid filled spaces bound by a membrane, which secrete and excrete wastes from the cell.	A. Cytoplasm B. Chloroplast C. Vacuoles D. Nucleus E. Centrioles F. Cell wall G. Mitochondria H. Cell membrane I. Ribosomes

LIST A	i	ii	iii	iv	v
LIST B					

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SECTION B (70 MARKS)

3. Give five difference between the cells of the rat and the cells of an onion

Rat's cells	Onion's cells
(i)	
(ii)	
(iii)	
(iv)	
(v)	

4. Briefly explain any five characteristics that can be observed in living things only.

(i) ……
……
(ii) ……
……..
(iii)……
……
(iv)……
……
(v) ……
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5. (a) What is photosynthesis?

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(b) Fill in the table below to briefly describe the epidermis and stomata as parts of the plant leaf:

	Description	Function
Epidermis		The functions of the epidermis are: protection against water loss by way of transpiration, regulation of gas exchange and secretion of metabolic compounds.
Stomata		

6. (a) State the cell theory.

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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(b)State the function of each of the following organelles:

(i) Nucleus

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(ii) Cell membrane

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(iii)Mitochondria

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(iv) Ribosomes

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(v) Vacuole

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(vi) Chloroplast

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7. (a) Binominal nomenclature is a system of naming whereby each organism is assigned two names; the first name being the Genus and the second name, the species. Given that the genus of humans is 'homo' and the species is 'sapiens', explain the rules that must be adhered to when using binomial nomenclature to name organisms using humans for examples.

(i)

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(ii)

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(iii)

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(b) Giving four points, briefly explain why artificial classification is not generally used when classifying organisms.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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8. Seypalm advised all Tanzanian students from form one to form four to study Biology. Give five points to explain why he gave them such advice.
   (i) ……
   ……
   (ii) ……
   ……
   (iii)……
   ……
   (iv)……
   ……

9. Many science fields apply the knowledge of biology. Outline five common fields that require the studying of biology.
   (i) ……
   ……
   (ii) ……
   ……..
   (iii)……
   ……
   (iv)……
   ……
   (v) ……
   ……

SECTION C (15 MARKS)

10. Biology is divided in to two main branches and many other smaller branches. Educate the public about the two main branches and any other six smaller branchea.

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